Multi-currency — income that arrives in USD and USDT

Epic: EPIC-013 · **Requirements:** A-04, A-14, A-15, A-06 · **Decisions:** ADR-0003 money and FX, schema §4

> **Status:** the data model and storage are merged (M2, ADR-0003). The entry UI (#174), the confirm-estimate flow (#177), the last-used rate (#179) and crypto year-to-date (#180) are in progress. The owner updates this section and adds screenshots when they merge.

Problem

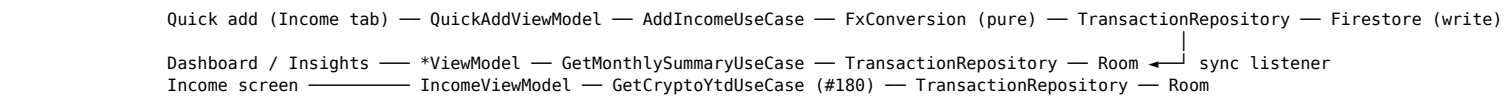
Kavindu's earlier tracker was "USD only" (**A-04**). His AdSense arrives in USD at a rate that changes every month, and he never recorded it (**A-14**). His Binance results are in USDT and can be losses (**A-15**). Income comes from several sources, each with its own currency (**A-06**). He needs every figure in LKR without any history changing when rates move.

Design decision

- **The LKR amount is the truth.** Every transaction stores `amountLkrMinor`, the LKR actually received, in minor units (`Long`). An optional `original` block stores the original currency, the amount in its minor units, and the rate as `fxRateMicros` (rate × 10^6, an integer).
- **The rate is locked at entry.** Totals, summaries and the goal only ever read `amountLkrMinor`. History is never recalculated with a newer rate (ADR-0003). Correcting an estimate edits that one entry only (#177).
- **Supported currencies:** LKR, USD and USDT. The Firestore rules enforce this set (`validOriginal`).
- **Two ways to enter a foreign amount** (#174):
  - type the **LKR received**, and the rate is derived; or
  - type the **rate**, and the LKR value is derived.

  Both go through a domain `FxConversion` helper in `BigDecimal` with HALF_UP, never `Double`.
- **Estimates:** an AdSense figure entered before the payout is known is an estimate. When the actual LKR arrives, editing it with "LKR received" makes it ACTUAL (#177). Until #177 lands, the demo seeder stores the estimate as `EXPECTED`.
- **Crypto results are signed.** A realised gain is positive and a realised loss is negative; crypto income is the only income allowed below zero. Moving money into or out of Binance is a **transfer**, never income (A-21).
- **Last-used rate per currency** goes in Preferences DataStore (#179). It is small, non-sensitive settings data that needs no sync or query, so it doesn't belong in Room or Firestore. The store keys one `fx_rate_<CURRENCY>` entry per currency, so a new currency needs no migration; the rate prefills Rate mode labelled "last used" and stays editable, and it is written only after a successful non-LKR income save, never from an abandoned draft. The rate a transaction keeps is still its own locked `fxRateMicros` — this store is a typing convenience, never read back into a stored amount.

Data flow



Edge cases handled

Case	Behaviour
Rate changes next month	Earlier entries keep their locked `amountLkrMinor`; nothing is recomputed.
A crypto loss	Negative `amountLkrMinor` and negative original amount; validation allows a negative only for crypto income.
Rounding	Half-cent results round HALF_UP once, at entry. For example, USD 48.20 × 298.40 = LKR 14,382.88.
An LKR entry	No `original` block; the rate is implicitly 1.0 and the status ACTUAL (#174 AC).
A foreign-currency invoice settles	It must carry the original amount and rate in the same currency as the invoice (`InvoiceRepositoryImpl.markReceived`).

Screenshots

TODO(M4) img/multi-currency-income-entry.png — Quick add, Income tab with USD
Quick add, Income: an AdSense entry in USD with its locked rate. Captured on seeded data once #174 merges. Until then the figure is the S2 wireframe (S2-quick-add.md).

Tests

- `DtoMappingTest` `OriginalAmount`: the map and round-trip.
- `TransactionTest`: crypto sign rules.
- #174: `FxConversion` tests (52.30 USD × 297.45 = LKR 15,556.64; rate derived from LKR; half-cent rounding).
- #180: year-to-date tests.